

Keith Chen

SOFTWARE DEVELOPER

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Profile

I am an experienced software engineer, mainly working on the backend area, programming in Python and Rust.

I have experience in developing web services, designing RESTful APIs, as well as collaborating with other team members.

Personally, I care about unit testing, documentation, and writing readable and testable code.

Projects

Nushell

April 2022 — Present

Nushell is a new type of shell that offers several features, including:

1. Multi-platform support: Nushell works natively on Linux, macOS, and Windows.
2. Structured data: Similar to PowerShell, everything in Nushell is structured data, so there is no need to treat everything as a bare string.
3. Powerful plugin system: Nushell provides a plugin system that is easy to use and extend.
4. Easy-to-use Nu language.

As a core team member, I have done the following:

1. Finished missing unit tests, making it easier for users to change their code with confidence.
2. Solved several bugs in Nushell, primarily to improve its compatibility with external commands.
3. Investigated and developed an Nushell library that empowers background jobs.
4. Added signature information to help users when they want to use a specific command, making commands much easier to use.
5. Make nushell works better with file redirection.
6. Reviewed pull requests from other contributors.

All my contributions can see here:

<https://github.com/nushell/nushell/pulls?q=author%3AWindSoilder>

Employment History

Software Engineer, Ricequant, Shenzhen

April 2018 — Present

1. I have prompted and rewritten a new C-S-based MongoDB synchronizer that enables syncing of internal MongoDB data to customers. Users can specify which database and collection to sync.

After careful design, the synchronizer now achieves the following:

- During full client synchronization, the original database can continue providing services as usual (without a direct increase in workload).
- Customers receive reliable and real-time updates from the original database.

- The company now requires fewer MongoDB instances (from 4 to 1).
2. I have redesigned and rewritten an web service named "Performance Attribution" for customers. This achieves the following:
- requiring significantly less memory (from 32GB to 8GB)
 - performing much faster (with a 70% reduction in runtime in general scenarios).
 - Additionally, the service is now more extendable and easier to test. I am also maintaining my company's stock data services
3. I'm also maintaining my company's data services, which includes:
- client side: stock data SDK for customers fetching financial data and market data easily
 - server side:
 - a. data server
 - b. bar-gateway server
 - c. index market data real time generator
 - d. user service
 - e. license service
 - f. etc..

More detailed: <https://www.ricequant.com/doc/rqdata/python/>

4. I am also working on the following:
- Developing finance services
 - Designing RESTful APIs
 - Making profile and performance improvements to existing code bases.
 - Rewriting some components in rust, it makes these components run 3.5x faster in general.

Tools and Technologies used include: Python 3, Rust, RESTful API design, distributed computing, MongoDB, Jira, Jenkins, Confluence, Redis

Associate Software Developer, Dell software, Zhuhai

June 2016 — April 2018

Initially, I used Python and Robot Framework to develop UAT automation testing for the product. I also worked on improving the performance of the testing code, which reduced the running time from 2 hours to 1 hour.

Later, I was involved in product development, where I mainly used C# and PowerShell.

Tools and technologies used included Python, Robot Framework, test automation, Agile methodologies, C#, AWS, Confluence, Jira, and Jenkins.

Links

[github](#) [blog](#)

Skills

MongoDB



Python



Rust



Education

Bachelor, Guangdong University of Technology, Guangzhou

September 2012 — June 2016

Internships

Test Development Engineer, Alibaba, Hangzhou

July 2015 — August 2015

Mainly working on testing security product.

The test process includes: Functional Testing, Performance Testing, Stability Testing

Tools & Technologies:

Python, test automation